

# DMS613: Introduction to Mathematical Finance

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Stochastic Calculus

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## 1 Random walk

### 1.1 Symmetric Random Walk

We model repeated fair coin tosses by an infinite sequence  $\omega = \omega_1\omega_2\cdots$ , where  $\omega_j \in \{H, T\}$  and  $\mathbb{P}(H) = \mathbb{P}(T) = \frac{1}{2}$ . Define

$$X_j(\omega) = \begin{cases} +1, & \omega_j = H, \\ -1, & \omega_j = T, \end{cases} \quad j \geq 1,$$

and set  $M_0 = 0$  and

$$M_k = \sum_{j=1}^k X_j, \quad k \geq 1.$$

The process  $(M_k)_{k \geq 0}$  is called the **symmetric random walk**. At each step it moves up or down by one unit, each with probability  $\frac{1}{2}$ .

For integers  $0 \leq k < \ell$ ,

$$M_\ell - M_k = \sum_{j=k+1}^{\ell} X_j.$$

Because the  $X_j$  are independent, increments over disjoint time intervals are independent. Moreover,

$$\mathbb{E}(M_\ell - M_k) = 0, \quad \text{Var}(M_\ell - M_k) = \sum_{j=k+1}^{\ell} \text{Var}(X_j) = \ell - k.$$

### 1.2 Martingale Property

Let  $\mathcal{F}_k = \sigma(X_1, \dots, X_k)$  be the information after  $k$  tosses. Then for  $\ell > k$ ,

$$\mathbb{E}[M_\ell \mid \mathcal{F}_k] = \mathbb{E}[(M_\ell - M_k) + M_k \mid \mathcal{F}_k] = \mathbb{E}[M_\ell - M_k] + M_k = M_k,$$

since  $M_\ell - M_k$  depends only on future tosses and has mean 0. Hence  $(M_k, \mathcal{F}_k)$  is a martingale.

### 1.3 Quadratic Variation

Define the quadratic variation up to time  $k$  by

$$[M, M]_k = \sum_{j=1}^k (M_j - M_{j-1})^2.$$

Here  $M_j - M_{j-1} = X_j \in \{\pm 1\}$ , so  $(M_j - M_{j-1})^2 = 1$  and therefore

$$[M, M]_k = \sum_{j=1}^k 1 = k.$$

This is a pathwise computation: along every realized path, the quadratic variation equals the elapsed time.

## 1.4 The natural filtration of the random walk

Recall that the underlying sample space is

$$\Omega = \{H, T\}^{\mathbb{N}} = \{\omega = (\omega_1, \omega_2, \dots) : \omega_j \in \{H, T\}\},$$

and that  $X_j(\omega) = +1$  if  $\omega_j = H$  and  $-1$  if  $\omega_j = T$ .

For each  $k \geq 0$ , define

$$\mathcal{F}_k = \sigma(X_1, \dots, X_k) = \sigma(M_0, M_1, \dots, M_k).$$

This is called the *natural filtration* of the random walk.

The  $\sigma$ -algebra  $\mathcal{F}_k$  represents all information revealed by the first  $k$  coin tosses.

$$\{\omega : X_1(\omega) = \varepsilon_1, \dots, X_k(\omega) = \varepsilon_k\}, \quad \varepsilon_j \in \{-1, +1\}.$$

Every event in  $\mathcal{F}_k$  is a union of such sets.

**Example 1.1** (The first two  $\sigma$ -algebras). For  $k = 1$ ,

$$\mathcal{F}_1 = \{\emptyset, \{X_1 = +1\}, \{X_1 = -1\}, \Omega\} = \{\emptyset, \{M_1 = 1\}, \{M_1 = -1\}, \Omega\}.$$

For  $k = 2$ , the four sets are

$$\begin{aligned} A_{++} &= \{X_1 = +1, X_2 = +1\} = \{M_1 = 1, M_2 = 2\}, \\ A_{+-} &= \{X_1 = +1, X_2 = -1\} = \{M_1 = 1, M_2 = 0\}, \\ A_{-+} &= \{X_1 = -1, X_2 = +1\} = \{M_1 = -1, M_2 = 0\}, \\ A_{--} &= \{X_1 = -1, X_2 = -1\} = \{M_1 = -1, M_2 = -2\}. \end{aligned}$$

Thus

$$\mathcal{F}_2 = \{\text{all unions of } A_{++}, A_{+-}, A_{-+}, A_{--}\},$$

which has  $2^4 = 16$  elements.

Moreover,

$$\{X_1 = +1\} = A_{++} \cup A_{+-} \in \mathcal{F}_2, \quad \{X_1 = -1\} = A_{-+} \cup A_{--} \in \mathcal{F}_2,$$

so  $\mathcal{F}_1 \subset \mathcal{F}_2$ . In general,

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \dots,$$

and  $(\mathcal{F}_k)_{k \geq 0}$  is a discrete-time filtration.

**Remark 1.** An event lies in  $\mathcal{F}_k$  if and only if it can be decided by observing the random walk up to time  $k$ . Events depending on future steps, such as  $\{M_{k+1} = M_k + 1\}$  or “the walk ever hits  $+10$ ”, are not in  $\mathcal{F}_k$ .

## 1.5 Scaled Random Walk

To approximate Brownian motion, we speed up time and scale down space. Fix  $n \in \mathbb{N}$  and define

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{\lfloor nt \rfloor}, \quad t \geq 0,$$

with linear interpolation between grid points. Then for  $0 \leq s < t$  with  $ns, nt \in \mathbb{N}$ ,

$$\mathbb{E}(W^{(n)}(t) - W^{(n)}(s)) = 0, \quad \text{Var}(W^{(n)}(t) - W^{(n)}(s)) = t - s.$$

Moreover, increments over disjoint intervals are independent. Thus the scaled walk has the same mean–variance structure as Brownian increments.

Its quadratic variation satisfies, pathwise,

$$[W^{(n)}, W^{(n)}](t) = \sum_{j=1}^{\lfloor nt \rfloor} \left( W^{(n)}\left(\frac{j}{n}\right) - W^{(n)}\left(\frac{j-1}{n}\right) \right)^2 = \sum_{j=1}^{\lfloor nt \rfloor} \frac{1}{n} \approx t.$$

## 1.6 Central Limit and the Brownian Limit

For fixed  $t \geq 0$ ,

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} \sum_{j=1}^{\lfloor nt \rfloor} X_j.$$

By the Central Limit Theorem,

$$W^{(n)}(t) \xrightarrow{d} \mathcal{N}(0, t).$$

More generally, the finite-dimensional distributions of  $\{W^{(n)}(t)\}_{t \geq 0}$  converge to those of a process with independent, stationary Normal increments of variance  $t - s$  over  $[s, t]$ .

## 2 Quadratic Variation

Fix a time horizon  $T > 0$  and consider a partition,

$$\Pi = \{0 = t_0 < t_1 < \cdots < t_n = T\}, \quad \|\Pi\| = \max_{0 \leq j \leq n-1} (t_{j+1} - t_j).$$

### 2.1 Quadratic variation: deterministic functions

**Definition 2.1** (Quadratic variation). *Let  $f : [0, T] \rightarrow \mathbb{R}$ . The quadratic variation of  $f$  up to time  $T$  is defined by*

$$[f, f](T) = \lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2,$$

*provided the limit exists.*

**Remark 2.** *Suppose  $f$  has a continuous derivative on  $[0, T]$ . By the mean value theorem,*

$$f(t_{j+1}) - f(t_j) = f'(t_j^*)(t_{j+1} - t_j) \quad \text{for some } t_j^* \in (t_j, t_{j+1}).$$

*Hence*

$$\sum_{j=0}^{n-1} (f(t_{j+1}) - f(t_j))^2 = \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j)^2 \leq \|\Pi\| \sum_{j=0}^{n-1} |f'(t_j^*)|^2 (t_{j+1} - t_j).$$

As  $\|\Pi\| \rightarrow 0$ , the second sum converges to  $\int_0^T |f'(t)|^2 dt$  (a Riemann sum), which is finite by continuity of  $f'$ . Therefore

$$[f, f](T) \leq \lim_{\|\Pi\| \rightarrow 0} \|\Pi\| \int_0^T |f'(t)|^2 dt = 0.$$

So continuously differentiable functions have zero quadratic variation.

### 3 Brownian Motion

**Definition 3.1.**  $W_t$ , a **Brownian motion**, is a continuous function in  $[0, T]$  where,

1.  $W_0 = 0$
2.  $W_t$  has independent increments, i.e.,  $W_t - W_s$  is independent of  $W_r - W_q$ , where  $0 \leq q \leq r \leq s \leq t$ .
3.  $W_t - W_s \sim N(0, t - s), t > s$ .

#### 3.1 Filtration for Brownian motion

**Definition 3.2** (Filtration for Brownian motion). Let  $(\Omega, \mathcal{F}, \mathbb{P})$  be a probability space on which a Brownian motion  $W = (W(t))_{t \geq 0}$  is defined. A filtration  $(\mathcal{F}(t))_{t \geq 0}$  for  $W$  is a collection of  $\sigma$ -algebras satisfying:

1. (**Information accumulates**) If  $0 \leq s < t$ , then  $\mathcal{F}(s) \subseteq \mathcal{F}(t)$ .
2. (**Adaptedness**) For each  $t \geq 0$ ,  $W(t)$  is  $\mathcal{F}(t)$ -measurable.
3. (**Independence of future increments**) For  $0 \leq t < u$ , the increment  $W(u) - W(t)$  is independent of  $\mathcal{F}(t)$ .

**Definition 3.3** (Adapted process). A stochastic process  $\Delta = (\Delta(t))_{t \geq 0}$  is said to be **adapted** to  $(\mathcal{F}(t))_{t \geq 0}$  if for each  $t \geq 0$ , the random variable  $\Delta(t)$  is  $\mathcal{F}(t)$ -measurable.

#### 3.2 Martingale property

**Theorem 3.1** (Brownian motion is a martingale). Let  $W$  be a Brownian motion with respect to  $(\mathcal{F}(t))_{t \geq 0}$ . Then for  $0 \leq s \leq t$ ,

$$\mathbb{E}[W(t) \mid \mathcal{F}(s)] = W(s).$$

*Proof.* Left as an exercise. □

#### 3.3 Quadratic variation of Brownian motion

**Definition 3.4.** Given a partition  $\Pi = \{0 = t_0 < \dots < t_n = T\}$ , define the sampled quadratic variation,

$$Q_\Pi(W; T) = \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2.$$

**Theorem 3.2** (Quadratic variation of Brownian motion). Let  $W_t$  be a Brownian motion. Then for every  $T \geq 0$ ,

$$[W, W](T) = \lim_{\|\Pi\| \rightarrow 0} \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2 = T \quad \text{almost surely.}$$

*Proof.* The increments  $\Delta_j W = W(t_{j+1}) - W(t_j)$  are independent and

$$\Delta_j W \sim \mathcal{N}(0, t_{j+1} - t_j).$$

Hence

$$\mathbb{E}[(\Delta_j W)^2] = t_{j+1} - t_j, \quad \mathbb{E}[(\Delta_j W)^4] = 3(t_{j+1} - t_j)^2,$$

(the fourth moment of a centered normal equals 3 times the variance squared). Therefore

$$\mathbb{E}[Q_\Pi] = \sum_{j=0}^{n-1} \mathbb{E}[(\Delta_j W)^2] = \sum_{j=0}^{n-1} (t_{j+1} - t_j) = T.$$

Next compute the variance. Since the summands are independent,

$$\text{Var}(Q_\Pi) = \sum_{j=0}^{n-1} \text{Var}((\Delta_j W)^2).$$

Using

$$\text{Var}((\Delta_j W)^2) = \mathbb{E}[(\Delta_j W)^4] - (\mathbb{E}[(\Delta_j W)^2])^2 = 3(t_{j+1} - t_j)^2 - (t_{j+1} - t_j)^2 = 2(t_{j+1} - t_j)^2,$$

we obtain

$$\text{Var}(Q_\Pi) = 2 \sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 \leq 2 \|\Pi\| \sum_{j=0}^{n-1} (t_{j+1} - t_j) = 2 \|\Pi\| T.$$

Hence  $\text{Var}(Q_\Pi) \rightarrow 0$  as  $\|\Pi\| \rightarrow 0$ . Since

$$\mathbb{E}[(Q_\Pi - T)^2] = \text{Var}(Q_\Pi),$$

we conclude that  $Q_\Pi \rightarrow T$  in  $L^2$ . This proves the quadratic-variation limit.  $\square$

**Remark 3.** Unlike  $C^1$  functions (which have zero quadratic variation), Brownian motion accumulates quadratic variation at rate one per unit time. This is a fundamental reason why stochastic calculus differs from ordinary calculus, and it is the mathematical source of volatility in diffusion models for asset prices.

**Remark 4.** One as a mnemonic writes  $dW(t)dW(t) = dt$ , meaning that when one computes quadratic variation, the squared increment behaves like time,  $\sum (\Delta W)^2 \approx \sum \Delta t = T$ . This identity should not be interpreted as an ordinary product of differentials, it is a shorthand for the quadratic variation limit above.

## Exercises

1. As  $\|\Pi\| \rightarrow 0$ , show that,  $\sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))(t_{j+1} - t_j) \rightarrow 0$ .
2. As  $\|\Pi\| \rightarrow 0$ , show that,  $\sum_{j=0}^{n-1} (t_{j+1} - t_j)^2 \rightarrow 0$ .

**Remark 5.** In the sense mentioned above, one therefore obtains mnemonics of the kind  $dW(t)dt = 0$  and  $dt dt = 0$ .

### 3.4 Volatility of Geometric Brownian Motion

Let  $\alpha \in \mathbb{R}$  and  $\sigma > 0$  be constants. Define the **geometric Brownian motion**

$$S(t) = S(0) \exp \left\{ \sigma W(t) + \left( \alpha - \frac{1}{2} \sigma^2 \right) t \right\}, \quad t \geq 0.$$

This is the asset-price model used in the Black–Scholes–Merton framework.

Fix  $0 \leq T_1 < T_2$  and suppose we observe  $S(t)$  for  $t \in [T_1, T_2]$ . Choose a partition

$$T_1 = t_0 < t_1 < \cdots < t_m = T_2,$$

and consider the log-returns,

$$\log \frac{S(t_{j+1})}{S(t_j)} = \sigma (W(t_{j+1}) - W(t_j)) + \left( \alpha - \frac{1}{2} \sigma^2 \right) (t_{j+1} - t_j), \quad j = 0, \dots, m-1.$$

The sum of squared log-returns (often called the realized volatility, different from implied volatility) is

$$\begin{aligned} \sum_{j=0}^{m-1} \left( \log \frac{S(t_{j+1})}{S(t_j)} \right)^2 &= \sigma^2 \sum_{j=0}^{m-1} (W(t_{j+1}) - W(t_j))^2 \\ &\quad + \left( \alpha - \frac{1}{2} \sigma^2 \right)^2 \sum_{j=0}^{m-1} (t_{j+1} - t_j)^2 \\ &\quad + 2\sigma \left( \alpha - \frac{1}{2} \sigma^2 \right) \sum_{j=0}^{m-1} (W(t_{j+1}) - W(t_j))(t_{j+1} - t_j). \end{aligned}$$

As  $\|\Pi\| \rightarrow 0$ ,

$$\begin{aligned} \sum_{j=0}^{m-1} (W(t_{j+1}) - W(t_j))^2 &\longrightarrow T_2 - T_1, \\ \sum_{j=0}^{m-1} (t_{j+1} - t_j)^2 &\longrightarrow 0, \quad \sum_{j=0}^{m-1} (W(t_{j+1}) - W(t_j))(t_{j+1} - t_j) \longrightarrow 0. \end{aligned}$$

Therefore, for fine partitions ( $\|\Pi\| \rightarrow 0$ ),

$$\sum_{j=0}^{m-1} \left( \log \frac{S(t_{j+1})}{S(t_j)} \right)^2 \approx \sigma^2 (T_2 - T_1),$$

or equivalently,

$$\frac{1}{T_2 - T_1} \sum_{j=0}^{m-1} \left( \log \frac{S(t_{j+1})}{S(t_j)} \right)^2 \approx \sigma^2. \quad (*)$$

### 3.5 Conditional expectation with Brownian motion

Let  $(W_t)_{t \geq 0}$  be a standard Brownian motion on  $(\Omega, \mathcal{F}, \mathbb{P})$  with its natural filtration  $\mathcal{F}_t = \sigma(W_u : 0 \leq u \leq t)$ . Fix  $0 \leq s \leq t$ .

## Exercises

1. Find  $\mathbb{E}[aW_t + b|\mathcal{F}_s]$ .
2. Find  $\mathbb{E}[W_s W_t|\mathcal{F}_s]$ .
3. Let  $0 \leq r \leq s \leq t$ , find  $\mathbb{E}[\mathbb{E}[W_t^2|\mathcal{F}_s]|\mathcal{F}_r]$ .
4. Let  $Z_t = \exp(\sigma W_t - \frac{1}{2}\sigma^2 t)$ . Show that  $Z_t$  is a martingale.  $Z_t$  is said to be an exponential martingale.

## 4 Trading gains and the Itô integral

Fix a time horizon  $T > 0$  and a partition (trading dates)

$$\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}.$$

A trader chooses, at each trading time  $t_j$ , a position  $\Delta(t_j)$  (number of shares) and then holds it constant until the next trading time and then sell it off.

Formally, we model the position process by a **simple adapted process**:

$$\Delta(t) = \sum_{j=0}^{n-1} \Delta(t_j) \mathbf{1}_{[t_j, t_{j+1})}(t),$$

where each  $\Delta(t_j)$  is  $\mathcal{F}_{t_j}$ -measurable (i.e. depends only on information available up to time  $t_j$ ). This is the non-anticipativity condition that you cannot choose your position using future price increments.

Consider what happens on a single interval  $[t_j, t_{j+1})$ . If you hold  $\Delta(t_j)$  shares throughout this interval, then your gain over the interval is

$$\text{gain on } [t_j, t_{j+1}) = \Delta(t_j)(W(t_{j+1}) - W(t_j)).$$

Summing over intervals, the total gain from 0 to  $T$  is

$$G_T(\Delta; \Pi) = \sum_{j=0}^{n-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)).$$

More generally, for an intermediate time  $t \in [t_k, t_{k+1})$ , the gain up to time  $t$  is

$$G_t(\Delta; \Pi) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)).$$

The choice  $\Delta(t_j)$  at time  $t_j$  must be based on  $\mathcal{F}_{t_j}$ , hence it must be independent of the future increment  $W(t_{j+1}) - W(t_j)$ . This matches the economic interpretation: you decide how many shares to hold before the price move on  $(t_j, t_{j+1}]$  is revealed.

This is exactly why the sum uses the left endpoint  $t_j$ . If we instead used  $\Delta(t_{j+1})$  (a right endpoint), then the strategy could depend on  $W(t_{j+1})$ , i.e. it could “peek into the future”, which is not an admissible trading rule.

**Definition 4.1** (Itô integral for a simple process). *Let  $\Delta$  be a simple adapted process of the form*

$$\Delta(t) = \sum_{j=0}^{n-1} \Delta(t_j) \mathbf{1}_{[t_j, t_{j+1})}(t).$$

We define its Itô integral with respect to Brownian motion by

$$\int_0^T \Delta(t) dW(t) = \sum_{j=0}^{n-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)),$$

and for  $t \in [t_k, t_{k+1})$ ,

$$\int_0^t \Delta(u) dW(u) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)).$$

**Remark 6** (Interpretation). The process  $\left(\int_0^t \Delta(u) dW(u)\right)_{0 \leq t \leq T}$  is exactly the cumulative trading gain when you hold  $\Delta(t)$  shares of the asset with (toy) price  $W(t)$ .

#### 4.1 The Itô integral is a martingale

Assume  $W$  is a Brownian motion with respect to  $(\mathcal{F}_t)_{t \geq 0}$  and  $\Delta$  is simple and adapted as above. Show that  $(I(t))_{0 \leq t \leq T}$  is a martingale, i.e. for  $0 \leq s \leq t \leq T$ ,

$$\mathbb{E}[I(t) \mid \mathcal{F}_s] = I(s).$$

*Proof.* Fix  $0 \leq s \leq t \leq T$ . Choose  $\ell, k$  such that  $s \in [t_\ell, t_{\ell+1})$  and  $t \in [t_k, t_{k+1})$ , with  $\ell \leq k$ . Using the representation (4.2.2), write

$$\begin{aligned} I(t) = & \underbrace{\sum_{j=0}^{\ell-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))}_{(A)} + \underbrace{\Delta(t_\ell)(W(s) - W(t_\ell))}_{(B)} \\ & + \underbrace{\Delta(t_\ell)(W(t_{\ell+1}) - W(s))}_{(C)} + \underbrace{\sum_{j=\ell+1}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j))}_{(D)} + \underbrace{\Delta(t_k)(W(t) - W(t_k))}_{(E)}. \end{aligned}$$

Now condition on  $\mathcal{F}_s$ .

Every term in (A) and (B) is  $\mathcal{F}_s$ -measurable, hence

$$\mathbb{E}[(A) + (B) \mid \mathcal{F}_s] = (A) + (B).$$

Finish the proof by showing  $\mathbb{E}[(C) + (D) + (E) \mid \mathcal{F}_s] = 0$  □

#### Exercises

1. Let  $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$  and let

$$\Delta(t) = \sum_{j=0}^{n-1} \Delta(t_j) \mathbf{1}_{[t_j, t_{j+1})}(t)$$

be a simple adapted process. For  $t \in [t_k, t_{k+1})$ , define

$$I(t) = \int_0^t \Delta(u) dW(u) = \sum_{j=0}^{k-1} \Delta(t_j)(W(t_{j+1}) - W(t_j)) + \Delta(t_k)(W(t) - W(t_k)).$$

Show that

$$\mathbb{E}[I(t)^2] = \mathbb{E}\left[\int_0^t \Delta(u)^2 du\right]. \quad (\text{Itô isometry})$$



2. Let  $\Delta$  be simple and adapted and let

$$I(t) = \int_0^t \Delta(u) dW(u).$$

Show that the quadratic variation of  $I$  up to time  $t$  satisfies

$$[I, I](t) = \int_0^t \Delta(u)^2 du, \quad 0 \leq t \leq T.$$

## 5 Itô Integral for General Integrands

We now extend the Itô integral from simple adapted integrands to more general stochastic processes.

**Definition 5.1.** A stochastic process  $\Delta(t)$ ,  $0 \leq t \leq T$ , is said to be an *admissible integrand* if:

1.  $\Delta(t)$  is adapted to  $\mathcal{F}(t)$

2.  $\mathbb{E} \left[ \int_0^T \Delta(t)^2 dt \right] < \infty.$

The second condition ensures that the total “risk exposure” of the strategy is finite.

Let  $\Delta(t)$  be an admissible integrand. We approximate  $\Delta$  by a sequence of simple adapted processes  $\{\Delta_n(t)\}_{n \geq 1}$  of the form

$$\Delta_n(t) = \sum_{j=0}^{k_n-1} \Delta(t_j^{(n)}) \mathbf{1}_{[t_j^{(n)}, t_{j+1}^{(n)})}(t),$$

where  $\Pi_n = \{0 = t_0^{(n)} < \dots < t_{k_n}^{(n)} = T\}$  is a partition of  $[0, T]$  and  $\|\Pi_n\| \rightarrow 0$ .

We require convergence in mean square:

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[ \int_0^T (\Delta_n(t) - \Delta(t))^2 dt \right] = 0$$

**Definition 5.2.** Let  $\Delta(t)$  be an admissible integrand and  $\Delta_n(t)$  a sequence of simple processes converging to  $\Delta$  in  $L^2([0, T] \times \Omega)$ . We define

$$\int_0^T \Delta(t) dW(t) = \lim_{n \rightarrow \infty} \int_0^T \Delta_n(t) dW(t),$$

where the limit is taken in  $L^2(\Omega)$ .

### 5.1 Basic properties

Let

$$I(t) = \int_0^t \Delta(u) dW(u).$$

Then the Itô integral satisfies:

1. **Adaptedness:**  $I(t)$  is  $\mathcal{F}(t)$ -measurable.
2. **Continuity:**  $t \mapsto I(t)$  has continuous sample paths.

3. **Martingale property:**

$$\mathbb{E}[I(t) \mid \mathcal{F}(s)] = I(s), \quad s \leq t.$$

4. **Quadratic Variation:**

$$[I, I](t) = \int_0^t \Delta^2(u) du$$

5. **Itô's Isometry:**

$$\mathbb{E}[(I(t))^2] = \mathbb{E} \left[ \int_0^t \Delta^2(u) du \right]$$

**Example 5.1** (Computing  $\int_0^T W(t) dW(t)$  via simple-process approximation). Fix  $T > 0$ . Let  $\Pi_n = \{t_j = jT/n : j = 0, 1, \dots, n\}$  be the uniform partition and define

$$W_j = W(t_j) = W\left(\frac{jT}{n}\right).$$

Approximate the integrand  $\Delta(t) = W(t)$  by the simple adapted process

$$\Delta_n(t) = W_j \quad \text{for } t_j \leq t < t_{j+1}, \quad j = 0, 1, \dots, n-1.$$

Then  $\Delta_n(t) \rightarrow W(t)$  in  $L^2([0, T] \times \Omega)$ , and by definition of the Itô integral,

$$\int_0^T W(t) dW(t) = \lim_{n \rightarrow \infty} \int_0^T \Delta_n(t) dW(t) = \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} W_j (W_{j+1} - W_j).$$

To simplify the sum, use the algebraic identity

$$W_{j+1}^2 - W_j^2 = (W_{j+1} - W_j)(W_{j+1} + W_j) = 2W_j(W_{j+1} - W_j) + (W_{j+1} - W_j)^2,$$

which rearranges to

$$W_j(W_{j+1} - W_j) = \frac{1}{2}(W_{j+1}^2 - W_j^2) - \frac{1}{2}(W_{j+1} - W_j)^2.$$

Summing from  $j = 0$  to  $n-1$  gives

$$\begin{aligned} \sum_{j=0}^{n-1} W_j(W_{j+1} - W_j) &= \frac{1}{2} \sum_{j=0}^{n-1} (W_{j+1}^2 - W_j^2) - \frac{1}{2} \sum_{j=0}^{n-1} (W_{j+1} - W_j)^2. \\ \sum_{j=0}^{n-1} (W_{j+1}^2 - W_j^2) &= W_n^2 - W_0^2 = W(T)^2 - W(0)^2 = W(T)^2, \end{aligned}$$

since  $W(0) = 0$ . Hence

$$\sum_{j=0}^{n-1} W_j(W_{j+1} - W_j) = \frac{1}{2}W(T)^2 - \frac{1}{2} \sum_{j=0}^{n-1} (W_{j+1} - W_j)^2.$$

As  $n \rightarrow \infty$ , the quadratic-variation limit yields

$$\sum_{j=0}^{n-1} (W_{j+1} - W_j)^2 \longrightarrow T$$

Taking limits, we conclude

$$\int_0^T W(t) dW(t) = \frac{1}{2}(W(T)^2 - T).$$

**Remark 7.** If  $g$  were differentiable, then  $\int_0^T g(t) dg(t) = \frac{1}{2}g(T)^2 - \frac{1}{2}g(0)^2$ . For Brownian motion, the extra term  $-\frac{1}{2}T$  appears because  $[W, W](T) = T$  (nonzero quadratic variation).

## 6 Itô's lemma

In ordinary calculus, if  $x(t)$  is differentiable and  $f \in C^2(\mathbb{R})$ , then

$$\frac{d}{dt}f(x(t)) = f'(x(t))x'(t), \quad \text{or} \quad df(x(t)) = f'(x(t))dx(t).$$

If we (incorrectly) tried to apply this to  $x(t) = W(t)$ , we would guess

$$df(W(t)) \stackrel{?}{=} f'(W(t))dW(t).$$

But Brownian motion is not differentiable and has nonzero quadratic variation:

$$[W, W](t) = t.$$

This produces an extra drift term in the correct chain rule.

**Theorem 6.1** (Itô's lemma: differential form). *Let  $W$  be a Brownian motion and let  $f \in C^2(\mathbb{R})$ . Then*

$$df(W(t)) = f'(W(t))dW(t) + \frac{1}{2}f''(W(t))dt.$$

**Theorem 6.2** (Itô's lemma: integral form). *Let  $W$  be a Brownian motion and let  $f \in C^2(\mathbb{R})$ . Then for  $t \geq 0$ ,*

$$f(W(t)) - f(W(0)) = \int_0^t f'(W(u))dW(u) + \frac{1}{2} \int_0^t f''(W(u))du.$$

**Theorem 6.3** (Itô's formula for Brownian motion). *Let  $(W(t))_{t \geq 0}$  be a Brownian motion with filtration  $(\mathcal{F}_t)$ . Let  $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$  be such that  $f_t, f_x, f_{xx}$  exist and are continuous (i.e.  $f \in C^{1,2}$ ). Then for every  $T \geq 0$ ,*

$$f(T, W(T)) = f(0, W(0)) + \int_0^T f_t(t, W(t))dt + \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2} \int_0^T f_{xx}(t, W(t))dt.$$

*Proof sketch.* Fix  $T > 0$ , and let  $\Pi = \{0 = t_0 < t_1 < \dots < t_n = T\}$  be a partition of  $[0, T]$ . Write the telescoping sum

$$f(T, W(T)) - f(0, W(0)) = \sum_{j=0}^{n-1} \left( f(t_{j+1}, W(t_{j+1})) - f(t_j, W(t_j)) \right).$$

For each  $j$ , apply the second-order Taylor expansion of  $f$  at  $(t_j, W(t_j))$  in the two variables  $(t, x)$ :

$$\begin{aligned} f(t_{j+1}, W(t_{j+1})) - f(t_j, W(t_j)) &= f_t(t_j, W(t_j))\Delta t_j + f_x(t_j, W(t_j))\Delta W_j \\ &\quad + \frac{1}{2}f_{xx}(t_j, W(t_j))(\Delta W_j)^2 + R_j, \end{aligned} \tag{1}$$

where  $\Delta t_j := t_{j+1} - t_j$ ,  $\Delta W_j := W(t_{j+1}) - W(t_j)$ , and  $R_j$  collects all the remaining terms:

$$R_j = f_{tx}(t_j, W(t_j))\Delta t_j \Delta W_j + \frac{1}{2}f_{tt}(t_j, W(t_j))(\Delta t_j)^2 + (\text{higher order terms}).$$

Summing (1) over  $j = 0, \dots, n-1$  gives

$$\begin{aligned} f(T, W(T)) - f(0, W(0)) &= \sum_{j=0}^{n-1} f_t(t_j, W(t_j))\Delta t_j + \sum_{j=0}^{n-1} f_x(t_j, W(t_j))\Delta W_j \\ &\quad + \frac{1}{2} \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j))(\Delta W_j)^2 + \sum_{j=0}^{n-1} R_j. \end{aligned} \tag{2}$$

By continuity of  $f_t(t, W(t))$ , the Riemann sum converges,

$$\begin{aligned}\sum_{j=0}^{n-1} f_t(t_j, W(t_j)) \Delta t_j &\xrightarrow{\Pi \rightarrow 0} \int_0^T f_t(t, W(t)) dt \\ \sum_{j=0}^{n-1} f_x(t_j, W(t_j)) \Delta W_j &\xrightarrow{\Pi \rightarrow 0} \int_0^T f_x(t, W(t)) dW(t).\end{aligned}$$

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) (\Delta W_j)^2 = \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) \Delta t_j + \sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) ((\Delta W_j)^2 - \Delta t_j).$$

The first sum is a Riemann sum,

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) \Delta t_j \xrightarrow{\Pi \rightarrow 0} \int_0^T f_{xx}(t, W(t)) dt.$$

For the second sum, use that  $(\Delta W_j)^2 - \Delta t_j$  has mean 0 and the increments are independent,

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) ((\Delta W_j)^2 - \Delta t_j) \xrightarrow{\Pi \rightarrow 0} 0,$$

Hence,

$$\sum_{j=0}^{n-1} f_{xx}(t_j, W(t_j)) (\Delta W_j)^2 \xrightarrow{\Pi \rightarrow 0} \int_0^T f_{xx}(t, W(t)) dt.$$

Taking limits in (2) yields

$$f(T, W(T)) - f(0, W(0)) = \int_0^T f_t(t, W(t)) dt + \int_0^T f_x(t, W(t)) dW(t) + \frac{1}{2} \int_0^T f_{xx}(t, W(t)) dt,$$

as claimed.  $\square$

**Example 6.1** (Compute  $\int_0^T W(t) dW(t)$  using Itô's formula). Let  $f(x) = x^2$ . Then  $f'(x) = 2x$  and  $f''(x) = 2$ . Applying Itô's lemma with this  $f$  gives

$$W(t)^2 - W(0)^2 = \int_0^t 2W(u) dW(u) + \frac{1}{2} \int_0^t 2 du.$$

Since  $W(0) = 0$ , this becomes

$$W(t)^2 = 2 \int_0^t W(u) dW(u) + t.$$

Rearranging,

$$\int_0^T W(u) dW(u) = \frac{1}{2} (W(T)^2 - T).$$

## 7 Itô Processes

**Definition 7.1** (Itô process). *Let  $W(t)$  be a Brownian motion with its natural filtration  $(\mathcal{F}_t)$ . A stochastic process  $X(t)$  is called an Itô process if it can be written as*

$$X(t) = X(0) + \int_0^t \Delta(u) dW(u) + \int_0^t \Theta(u) du,$$

where:

- $X(0)$  is deterministic,
- $\Delta(u)$  and  $\Theta(u)$  are adapted processes,
- $\mathbb{E}\left[\int_0^t \Delta^2(u) du\right] < \infty$  for all  $t$ .

Here  $\Delta(u)$  is called the volatility and  $\Theta(u)$  the drift.

**Lemma 7.1.** *Let  $X(t)$  be an Itô process of the form above. Then its quadratic variation satisfies*

$$[X, X](t) = \int_0^t \Delta^2(u) du.$$

**Remark 8.** *The finite-variation drift term  $\int_0^t \Theta(u) du$  contributes no quadratic variation. Only the stochastic integral with respect to  $W$  contributes.*

### 7.1 Itô Formula

**Theorem 7.2.** *Let  $X(t)$  be an Itô process and let  $f(t, x) \in C^{1,2}(\mathbb{R}_+ \times \mathbb{R})$ . Then*

$$df(t, X(t)) = f_t(t, X(t)) dt + f_x(t, X(t)) dX(t) + \frac{1}{2} f_{xx}(t, X(t)) d[X, X](t).$$

Substituting

$$dX(t) = \Delta(t) dW(t) + \Theta(t) dt, \quad d[X, X](t) = \Delta^2(t) dt,$$

we obtain,

$$\begin{aligned} df(t, X(t)) &= f_t(t, X(t)) dt + f_x(t, X(t)) \Delta(t) dW(t) \\ &\quad + f_x(t, X(t)) \Theta(t) dt + \frac{1}{2} f_{xx}(t, X(t)) \Delta^2(t) dt. \end{aligned}$$

### 7.2 Integral Form of Itô's Formula

Integrating from 0 to  $t$ , we obtain

$$\begin{aligned} f(t, X(t)) - f(0, X(0)) &= \int_0^t f_t(u, X(u)) du + \int_0^t f_x(u, X(u)) dX(u) \\ &\quad + \frac{1}{2} \int_0^t f_{xx}(u, X(u)) d[X, X](u). \end{aligned}$$

**Example 7.1** (Generalized Geometric Brownian Motion). *Let  $(W(t))_{t \geq 0}$  be a Brownian motion with filtration  $(\mathcal{F}_t)$ . Let  $\alpha(t)$  and  $\sigma(t)$  be adapted processes satisfying the usual integrability conditions.*

*Define the Itô process*

$$X(t) = \int_0^t \sigma(s) dW(s) + \int_0^t \left( \alpha(s) - \frac{1}{2} \sigma^2(s) \right) ds.$$

Equivalently, in differential form,

$$dX(t) = \sigma(t) dW(t) + \left( \alpha(t) - \frac{1}{2}\sigma^2(t) \right) dt.$$

Define the asset price process by

$$S(t) = S(0)e^{X(t)},$$

where  $S(0) > 0$  is deterministic.

Let

$$f(x) = S(0)e^x, \quad f'(x) = S(0)e^x, \quad f''(x) = S(0)e^x.$$

Then  $S(t) = f(X(t))$ , and we may apply the Itô's formula.

$$\begin{aligned} dS(t) &= df(X(t)) \\ &= f'(X(t)) dX(t) + \frac{1}{2}f''(X(t)) d[X, X](t). \end{aligned}$$

Since  $X(t)$  is an Itô process with volatility  $\sigma(t)$ ,

$$d[X, X](t) = \sigma^2(t) dt.$$

Substituting,

$$\begin{aligned} dS(t) &= S(0)e^{X(t)} dX(t) + \frac{1}{2}S(0)e^{X(t)}\sigma^2(t) dt \\ &= S(t) dX(t) + \frac{1}{2}S(t)\sigma^2(t) dt. \end{aligned}$$

Now substitute the expression for  $dX(t)$ :

$$\begin{aligned} dS(t) &= S(t) \left[ \sigma(t) dW(t) + \left( \alpha(t) - \frac{1}{2}\sigma^2(t) \right) dt \right] + \frac{1}{2}S(t)\sigma^2(t) dt \\ &= \alpha(t)S(t) dt + \sigma(t)S(t) dW(t). \end{aligned}$$

## Interpretation

The asset price  $S(t)$  satisfies the stochastic differential equation

$$dS(t) = \alpha(t)S(t) dt + \sigma(t)S(t) dW(t).$$

- $\alpha(t)$  is the instantaneous mean rate of return,
- $\sigma(t)$  is the instantaneous volatility.

Both coefficients may be time-varying and random.

## Itô integral with deterministic integrand is Normally Distributed

**Theorem 7.3** (Itô integral of a deterministic integrand). *Let  $(W_t)_{t \geq 0}$  be a Brownian motion and let  $\Delta : [0, \infty) \rightarrow \mathbb{R}$  be a deterministic function such that  $\int_0^t \Delta^2(s) ds < \infty$  for every  $t$ . Then for each  $t \geq 0$ ,*

$$I(t) \sim N\left(0, \int_0^t \Delta^2(s) ds\right),$$

**Remark 9.** Please refer to Shreve for a proof of this. To show mean is 0, we use that the Itô integral is a martingale and the variance follows from the quadratic variation of an Itô integral.

### 7.3 Example: Cox-Ingersoll-Ross (CIR) interest rate model

**Example 7.2.** Let  $(W_t)_{t \geq 0}$  be a Brownian motion. The Cox-Ingersoll-Ross (CIR) model for the (short) interest rate  $(R_t)_{t \geq 0}$  is

$$dR(t) = (\alpha - \beta R(t)) dt + \sigma \sqrt{R(t)} dW(t), \quad \alpha, \beta, \sigma > 0. \quad (3)$$

Consider the function  $f(t, x) = e^{\beta t} x$  and apply Itô's formula to  $f(t, R(t)) = e^{\beta t} R(t)$ . We have  $f_t(t, x) = \beta e^{\beta t} x$ ,  $f_x(t, x) = e^{\beta t}$ ,  $f_{xx}(t, x) = 0$ , hence

$$\begin{aligned} d(e^{\beta t} R(t)) &= f_t(t, R(t)) dt + f_x(t, R(t)) dR(t) + \frac{1}{2} f_{xx}(t, R(t)) dR(t) dR(t) \\ &= \beta e^{\beta t} R(t) dt + e^{\beta t} [(\alpha - \beta R(t)) dt + \sigma \sqrt{R(t)} dW(t)] \\ &= \alpha e^{\beta t} dt + \sigma e^{\beta t} \sqrt{R(t)} dW(t). \end{aligned} \quad (4)$$

Integrating from 0 to  $t$  gives

$$e^{\beta t} R(t) = R(0) + \alpha \int_0^t e^{\beta u} du + \sigma \int_0^t e^{\beta u} \sqrt{R(u)} dW(u) = R(0) + \frac{\alpha}{\beta} (e^{\beta t} - 1) + \sigma \int_0^t e^{\beta u} \sqrt{R(u)} dW(u).$$

Taking expectations,

$$e^{\beta t} \mathbb{E}[R(t)] = R(0) + \frac{\alpha}{\beta} (e^{\beta t} - 1),$$

and hence,

$$\mathbb{E}[R(t)] = e^{-\beta t} R(0) + \frac{\alpha}{\beta} (1 - e^{-\beta t}). \quad (5)$$

### Exercises

1. Find  $\mathbb{E}[W_a W_b]$ .
2. Let  $(W_t)_{t \geq 0}$  be a Brownian motion with its natural filtration  $(\mathcal{F}_t)_{t \geq 0}$ . Fix  $0 < s < t$ . Suppose you are given the entire path  $\{W(u) : 0 \leq u \leq s\}$ . For each of the following quantities, state whether it is already determined at time  $s$  (i.e.  $\mathcal{F}_s$ -measurable) or still random at time  $s$ . Justify your answer briefly.
  1.  $W_t - W_s$ ,
  2.  $\sup_{0 \leq u \leq t} W(u)$ ,
  3.  $\int_0^s W(u) du$ ,
  4.  $\int_s^t W(u) du$ .
3. Let  $(W_t)_{t \geq 0}$  be a Brownian motion with filtration  $(\mathcal{F}_t)$ . A trading strategy  $\Delta(t)$  is said to be admissible if it is adapted to  $(\mathcal{F}_t)$ .

Determine which of the following processes are admissible trading strategies.

1.  $\Delta(t) = \mathbf{1}_{\{W(t) > 0\}}$ ,
2.  $\Delta(t) = \mathbf{1}_{\{W(T) > 0\}}$ , where  $T > t$  is fixed,
3.  $\Delta(t) = \sup_{0 \leq u \leq t} W(u)$ ,

4.  $\Delta(t) = W(t + \varepsilon)$  for some fixed  $\varepsilon > 0$ .

4. It is known that the process

$$M_t = W_t^2 - t$$

is a martingale.

Now define

$$N_t = |W_t| - c\sqrt{t},$$

where  $c > 0$  is a constant.

Is it possible to choose  $c$  such that  $(N_t)_{t \geq 0}$  becomes a martingale?

5. A “too good to be true” trading strategy. Fix  $0 < s < t$  and consider the process

$$\Delta(u) = \text{sgn}(W_t - W_s) \mathbf{1}_{(s,t]}(u).$$

1. Is  $\Delta$  adapted? Explain.

2. If it were allowed, what would  $\int_s^t \Delta(u) dW(u)$  equal?

3. Why does this illustrate the role of adaptedness in the Itô integral?

6. Let  $0 \leq t \leq T$  and define

$$J_t = \int_0^t W_u du, \quad I_t = \int_0^t J_u dW_u.$$

1. Show that  $J_t$  is  $\mathcal{F}_t$ -measurable for each  $t$ .

2. Use Itô isometry to express  $\mathbb{E}[I_t^2]$  in terms of  $\mathbb{E}[J_u^2]$ .

3. Compute  $\mathbb{E}[J_u^2]$ .

7. Fix  $0 < s < t$  and consider the “right-endpoint” Riemann sum

$$R_n = \sum_{j=0}^{n-1} W(t_{j+1})(W(t_{j+1}) - W(t_j)), \quad t_j = \frac{js}{n}.$$

1. Show that  $R_n = \frac{1}{2}(W(s)^2 + \sum_{j=0}^{n-1} (W(t_{j+1}) - W(t_j))^2)$ .

2. Take limits and identify  $\lim_{n \rightarrow \infty} R_n$ .

3. Compare with  $\int_0^s W(u) dW(u)$  and explain the difference.

8. Define,

$$\tilde{W}_t = \frac{1}{\sqrt{c}} W_{ct}$$

1. Is  $\tilde{W}_t$  a Brownian motion?

2. What is the quadratic variation of  $\tilde{W}_t$ ?

9. Consider the Geometric Brownian Motion,

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

1. Find the distribution of  $S_t$ .

2. Find the mean and variance of  $S_t$ .

10. Find the Variance of the CIR process.



11. Find  $\int_0 W_t^2 dW_t$

12. Let  $X_t = \exp(W_t + \mu t)$ . Find  $d(\log(1 + X_t))$ .

13. Consider GBM,

$$dS_t = \mu S_t dt + \sigma S_t dB_t$$

Then show that the discounted process  $\tilde{S}_t = e^{-\mu t} S_t$  is a martingale.

14. Consider GBM as above. Compute  $f(p)$  such that,

$$M_t = e^{-f(p)t} S_t^p$$

is a martingale.